

BYTEFORCE

Smart Recruitment Matching Platform

HOW TO RUN THE PROJECT

Setup instructions, required login credentials, URLs, and verification steps

Project	SmartRecruitmentMatchingPlatform
Framework	.NET 8 / ASP.NET Core Web API
Database	SQL Server LocalDB - SmartRecruitmentDb
Frontend	HTML, CSS and JavaScript served from wwwroot

Document basis: This guide was prepared by inspecting the supplied SmartRecruitmentMatchingPlatform.zip, including Program.cs, appsettings.json, launchSettings.json, DatabaseSeeder.cs, migrations, and frontend files.

1. Prerequisites

Install or confirm the following on the Windows computer before running the application:

Requirement	What is required
Visual Studio Community	with the ASP.NET and web development workload. The project can be opened using SmartRecruitmentMatchingPlatform.slnx; if the installed Visual Studio does not support .slnx, open SmartRecruitmentMatchingPlatform.csproj directly.
.NET 8 SDK	required because the project targets net8.0.
SQL Server LocalDB	required by the configured connection string Server=(localdb)\MSSQLLocalDB. LocalDB is commonly installed with Visual Studio / SQL Server Express components.
Web browser	Microsoft Edge, Google Chrome, or another modern browser.
Internet connection for first restore	needed only if the required NuGet packages are not already cached on the computer.

2. Required Login Credentials

The project seeds one Administrator account automatically. Job Seeker and Employer accounts are not pre-seeded and must be registered from the user interface before they can log in.

Role	Email	Password	How account is created
Administrator	admin@smartrecruitment.com	Admin123	Automatically seeded on first startup
Job Seeker	No default credential	Create your own	Register at /auth/register-jobseeker.html
Employer	No default credential	Create your own	Register at /auth/register-employer.html

Important: Registration passwords must satisfy the backend rule: at least 8 characters and include an uppercase letter, a lowercase letter, and a number. Example valid test password: TestUser123.

3. Run the Project in Visual Studio - Recommended Method

1. Extract the ZIP file

Extract SmartRecruitmentMatchingPlatform.zip to a normal local folder. Avoid running the solution directly from inside the compressed ZIP.

2. Open the project

Open the extracted SmartRecruitmentMatchingPlatform folder. Double-click SmartRecruitmentMatchingPlatform.slnx, or open Visual Studio and select Open a project or solution. If .slnx is not supported, select SmartRecruitmentMatchingPlatform.csproj.

3. Restore NuGet packages

Visual Studio normally restores packages automatically. If dependencies appear missing, right-click the project and choose Restore NuGet Packages. The project uses ASP.NET Core JWT Bearer, Entity Framework Core SQL Server/Design/Tools, and Swashbuckle.

4. Confirm the database connection

Open appsettings.json and confirm the DefaultConnection still points to the configured LocalDB instance:

```
Server=(localdb)\MSSQLLocalDB;  
Database=SmartRecruitmentDb;  
Trusted_Connection=True; TrustServerCertificate=True;  
MultipleActiveResultSets=true
```

5. Choose the HTTPS project profile

In the Visual Studio run-profile selector, choose the project HTTPS profile (not a separate frontend server). The configured development URLs are https://localhost:7228 and http://localhost:5249.

6. Run the application

Press F5 to run with debugging, or Ctrl+F5 to run without debugging. On first run, allow the HTTPS development certificate prompt if Visual Studio displays one.

7. Allow automatic database setup

No manual migration command is required for the normal first run. Program.cs calls Database.Migrate(), which creates/updates SmartRecruitmentDb using the included EF Core migrations. DatabaseSeeder then adds the default skills and Administrator account if they do not already exist.

8. Open the application

The project profile initially launches Swagger. To use the full frontend, open the application root or the login page using the same HTTPS port.

4. Main Development URLs

Purpose	URL using the HTTPS profile
Home page	https://localhost:7228/
Login	https://localhost:7228/auth/login.html
Job Seeker registration	https://localhost:7228/auth/register-jobseeker.html
Employer registration	https://localhost:7228/auth/register-employer.html
Swagger / API testing	https://localhost:7228/swagger

If the port is different: Use the exact localhost address shown by Visual Studio or the browser. The code uses relative /api/... paths, so the frontend and backend work together from the same ASP.NET Core host.

5. First Login and Role Setup

Administrator

1. Go to https://localhost:7228/auth/login.html.
2. Enter admin@smartrecruitment.com and Admin123.
3. Select Login. The frontend reads the returned role and redirects the Administrator to /admin/dashboard.html.

Job Seeker

4. Open /auth/register-jobseeker.html.
5. Enter full name, email, password, and confirm password. Use at least 8 characters with uppercase, lowercase, and a number.
6. After successful registration, return to the login page and sign in using the account you just created.
7. Complete the profile, add skills, upload a CV if needed, browse jobs, and submit applications.

Employer

8. Open /auth/register-employer.html.
9. Enter full name, company name, email, password, and confirm password.
10. After registration, log in with the newly created Employer account.
11. Complete the employer profile and create vacancies before testing applicant and contact-request workflows.

CV storage: Uploaded CV files are stored locally under ProtectedFiles/CVs. The folder is created automatically by LocalFileStorageService. The configured maximum file size is 5 MB and allowed extensions are .pdf and .docx.

6. Quick End-to-End Verification

- ☐ Application starts without an unhandled error.
- ☐ Swagger loads at /swagger in Development mode.
- ☐ The home page and login page load with CSS and JavaScript correctly.
- ☐ Administrator can log in with the seeded credentials.

- ☐ A new Job Seeker can register and then log in.
- ☐ A new Employer can register and then log in.
- ☐ SmartRecruitmentDb exists in SQL Server LocalDB after first startup.
- ☐ Role-based dashboards open after login.

7. Alternative: Run from the .NET CLI

This is optional. It is useful if Visual Studio is unavailable but .NET 8 SDK and SQL Server LocalDB are installed.

```
cd <extracted-folder>\SmartRecruitmentMatchingPlatform
dotnet restore
dotnet run --launch-profile https
```

Then open <https://localhost:7228/> or <https://localhost:7228/swagger>. If the machine does not trust the development HTTPS certificate, run the following once in a terminal and accept the trust prompt:

```
dotnet dev-certs https --trust
```

8. Troubleshooting

Issue	Recommended fix
LocalDB connection error	Confirm SQL Server LocalDB is installed and the instance name is MSSQLLocalDB. The project expects Windows integrated authentication through Trusted_Connection=True.
Database schema error	Stop and restart the application first. Program.cs automatically applies all included migrations at startup. If the database was manually modified and is inconsistent, use SQL Server Object Explorer to inspect SmartRecruitmentDb before resetting it.
Admin login fails	Confirm the email is admin@smartrecruitment.com and password is Admin123. The admin is seeded only if that email does not already exist in the Users table.
Registration says password is invalid	Use 8+ characters with at least one uppercase letter, one lowercase letter, and one number. The server enforces this even if browser-side feedback is less strict.
HTTPS certificate warning	Use the Visual Studio-generated HTTPS certificate, or run dotnet dev-certs https --trust from a terminal.
Swagger opens but frontend was expected	Swagger is the launch URL in launchSettings.json. Manually change the browser path to /, /index.html, or /auth/login.html.
CSS/JavaScript appears missing	Do not open the HTML files directly with file://. Run the ASP.NET Core application and access pages through https://localhost:<port>/ so wwwroot static files and /api routes share the same host.
CV upload fails	Use only .pdf or .docx files and keep the file size at or below 5 MB.

9. Project Startup Summary

Component	What happens when the project runs
Backend	ASP.NET Core application starts and exposes REST API controllers.
Frontend	Static HTML/CSS/JavaScript is served by the same application from wwwroot.
Database	EF Core connects to SmartRecruitmentDb on (localdb)\MSSQLLocalDB.
Migrations	Applied automatically during startup through Database.Migrate().
Seed data	Default skills and the Administrator account are added when absent.
Authentication	Login returns a JWT token; the frontend stores the session in browser storage and redirects according to the user role.
CV files	Stored locally in ProtectedFiles/CVs; directory is automatically created.

Submission-ready credential statement: Administrator login: admin@smartrecruitment.com / Admin123. Employer and Job Seeker credentials are created through their registration pages because no fixed accounts for those roles are seeded in the supplied project.